



AJKD Atlas of Renal Pathology: Acute T-Cell-Mediated Rejection

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Clinical and Pathologic Features

Acute cellular rejection, also called acute T-cell-mediated rejection (TCMR), presents in the transplant recipient with acute kidney injury and decreased urine output, and may be accompanied by proteinuria. In more severe cases there can be transplant tenderness. TCMR can develop at any time, as early as a week or as late as years after transplantation. With current immunosuppressive therapy, acute rejection develops in about 10%-12% of transplant patients. Pure TCMR responds well to increased immunosuppressive therapy.

Light microscopy: Inflammation and tubulitis is scored. i1 is defined as inflammation involving 10%-25% of nonscarred parenchyma; i2 as 26%-50% involvement, and i3 as >50% involvement. Tubulitis is scored as t1 when 1-4 cells are present per tubular cross-section in non-atrophic tubules; t2 is 5-10 cells, and t3 is >10 cells.

Diagnostic criteria for TCMR according to Banff classification 2005 (Solez et al, *Am J Transplant* 2007;7(3):518-526) include the following:

- Type IA: significant interstitial inflammation (i2 or i3) and foci of moderate tubulitis (t2)
- Type IB: significant interstitial inflammation (i2 or i3) and foci of severe tubulitis (t3)
- Type IIA (mild vascular rejection): mild to moderate intimal arteritis, also called endothelialitis, involving <25% of the luminal circumference (v1)
- Type IIB (moderate vascular rejection): severe intimal arteritis involving ≥25% of the luminal area (v2)
- Type III (severe vascular rejection): transmural arteritis and/or arterial fibrinoid change and necrosis (v3)
- Types II and III frequently also have antibody-mediated components

Subtypes include plasma cell-rich cellular rejection and eosinophil-rich cellular rejection. However, the threshold for type I acute TCMR is high, and in clinical settings of presumed acute rejection, morphologic changes most often fall in the borderline category,

defined as foci of tubulitis with minor interstitial inflammation (Banff i0 or i1), or interstitial inflammation (i2 or i3) with minor tubulitis (t1). Therefore, Banff working groups are assessing lower thresholds of i and t lesions for diagnosis of TCMR. The Cooperative Clinical Trials in Transplantation (CCTT) criteria set the threshold for acute TCMR as >5% of nonscarred parenchyma with inflammation, and at least 3 non-scarred tubules with tubulitis in 10 high-power fields. Biopsy adequacy for CCTT specificity was set as 2 pieces of cortex with 7 glomeruli and 2 arteries.

Immunofluorescence microscopy: C4d is negative in peritubular capillaries in pure TCMR.

Electron microscopy: No specific changes.

Etiology/Pathogenesis

Acute TCMR is due to cytotoxic T-cell reaction to HLA antigens on the donor kidney. These T cells can affect the tubules, interstitium, arteries, and even glomeruli. Mixed acute antibody-mediated rejection and TCMR can occur.

Differential Diagnosis

Polyomavirus nephropathy causes interstitial inflammation and tubulitis, and is distinguished from type I TCMR by the pleomorphic infiltrate with lymphocytes, plasma cells, occasional polymorphonuclear neutrophils (PMNs) and eosinophils, viral cytopathic changes, and SV40 positivity by immunostaining. Acute tubulointerstitial nephritis due to drug hypersensitivity reaction may have more frequent eosinophils, but its appearance overlaps substantially with TCMR. Acute pyelonephritis has intratubular casts of PMNs and frequent PMNs in the interstitium. Acute antibody-mediated rejection may occur in combination with TCMR, and is diagnosed with positive serum donor-specific antibodies and positive C4d staining of peritubular capillaries and significant microcirculation inflammation (ie, inflammation in glomeruli and peritubular capillaries), and is suspected with type III changes described above.

Endothelialitis in the transplant is highly indicative of type II TCMR, but recurrent vasculitis may produce a similar lesion, but usually with necrosis. Acute thrombotic microangiopathy may have occasional inflammatory cells or even fibrinoid necrosis.

Key Diagnostic Features

- Type I: Interstitial inflammation in unscarred cortex and tubulitis
- Type II: Endothelialitis in artery/arteriole

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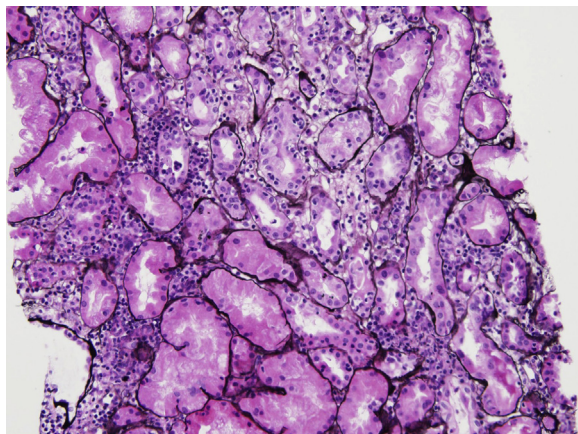


Figure 1. Acute cellular rejection (T-cell mediated, type I) with interstitial lymphocytic infiltrate, foci of moderate tubulitis, and interstitial edema (Jones silver stain).

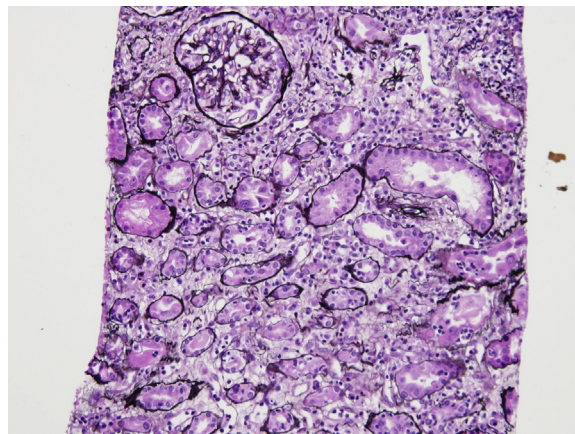


Figure 3. Acute cellular rejection (T-cell mediated, type I) with interstitial lymphocytic infiltrate, foci of mild tubulitis, and interstitial edema (Jones silver stain).

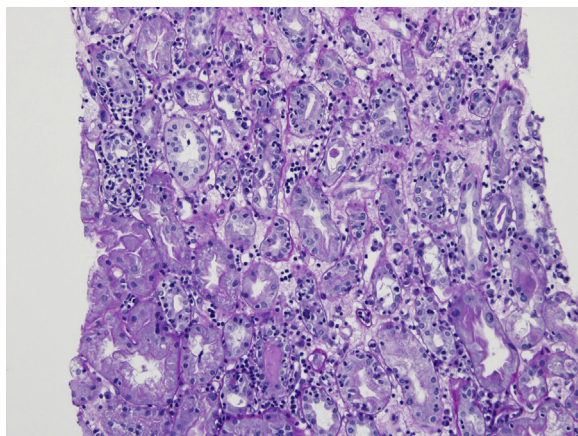


Figure 2. Acute cellular rejection (T-cell mediated, type I) with interstitial lymphocytic infiltrate, foci of severe tubulitis, and interstitial edema (periodic acid–Schiff stain).

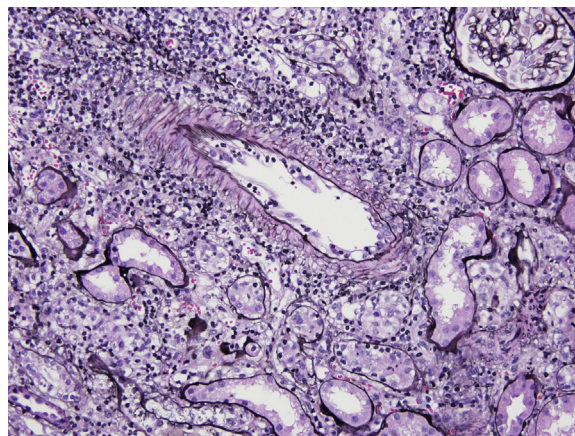


Figure 4. Acute cellular rejection (T-cell mediated, type II), also called acute vascular rejection, with interstitial lymphocytic infiltrate, foci of moderate tubulitis, and endothelialitis of an interlobular artery (Jones silver stain).